



Politecnico
di Torino

Algorithms and Programming

Iniziato	martedì, 21 dicembre 2021, 17:10
Stato	Completato
Terminato	martedì, 21 dicembre 2021, 17:10
Tempo impiegato	13 secondi
Valutazione	0,00 su un massimo di 14,00 (0%)

Domanda 1

Risposta non data

Punteggio max.:
2,00

The main program calls function f as f(3, 5). In turn function f calls function g.

```
void f (int n, int m) {
    printf ("F");
    if (n<=0)
        return;
    if (n%2==0) {
        g(n,m-1);
    } else {
        f(n-1,m);
    }
    return;
}

void g (int n, int m) {
    printf ("G");
    if (m<=0)
        return;
    if (m%2==0) {
        g(n,m-1);
    } else {
        f(n-1,m);
    }
    return;
}
```

Following the recursion tree of f and g, indicate the number of times the character 'F' and the character 'G' are displayed on standard output.

Please, report your response as a pair of integer values (the first one for 'F', the second one for 'G'). No other symbols must be included in the response. For example, if the function displays "FGFGFGGG", you must reply: 3 5

Risposta:



La risposta corretta è : 4 2

Domanda 2

Risposta non data

Punteggio max.:

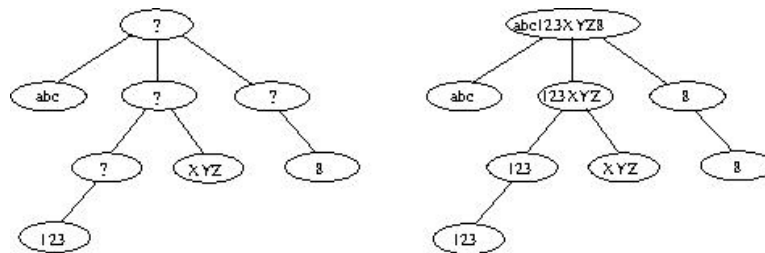
4,00

A n-ary tree is organized adopting the following C data structure:

```
typedef struct e_s e_t;
struct e_s {
    char *str;
    int n;
    e_t **children;
};
```

where **str** is the identifier of the node of the tree, **n** is the number of children of the node, and **children** is the array of pointers referring to these children.

At the very beginning, the field **str** is defined only for the leaves of the tree, as represented by the leftmost tree of the following picture. For all internal nodes, **str** is not NULL but it is undefined (i.e., it points to an undefined, dynamically allocated, and useless string).



Write the function

```
<type> tree_complete (e_t *root);
```

which receives the tree rooted at **root** and it recursively completes the tree by assigning to each field **str** (but the ones on the leaves that have to remain untouched) a string which is the concatenation of the strings stored in the node's children. The result of this operation for the leftmost tree of the picture should deliver the rightmost tree of the same picture. The **type** of the function **tree_complete**, i.e., **<type>**, can be selected by the candidate.

Please, report the entire C code of the function. Allocate and free all required dynamic fields avoiding memory leaks. Briefly explain (in plain English language, within a C comment) the logic of the code.

Domanda 3

Risposta non data

Punteggio max.:

8,00

A square matrix of size n stores only integer values.

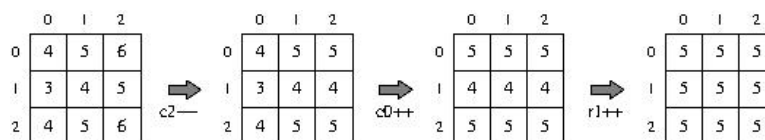
Suppose there are two commands that, given a row or a column index, are able to increment or decrement the value of all cells belonging to that specific row or column by 1. Thus the command **r0++** increases all elements of row 0 by 1, whereas **c3--** decreases all values of column 3 by 1.

Write the function:

```
void matrix_equalizer (int **mat, int n);
```

that receives the square matrix **mat** and its size **n**, and finds the shortest sequence of commands (if this sequence does exist) that can modify the matrix **mat** such that it contains the same integer value (whatever this value is) in all elements.

For example, give the matrix represented in the following picture on the left-hand side, the arrows represent 3 commands (given in sequence, i.e., **c2--**, **c0++**, and **r1++**) which generate the 3 matrices following the leftmost one. The final matrix does contain only the value 5. Thus the sequence of commands produces the desired result and it is the shortest sequence changing all elements of the matrix into the same value.



Notice that function **matrix_equalizer** may act as a wrapper and it can possibly call other recursive functions. Moreover, commands, e.g., **c2--**, **c0++**, etc., can be encoded as desired (i.e., as integer values or code strings).

Please, report the entire C code of the function. Briefly explain (in plain English language, within a C comment) the logic of the code and the combinatorics principle that has been adopted.